Last Updated: October 2026

Shuhaib Mehri

shuhaibm.github.io
mehri2@illinois.edu / mesm.shuhaib@gmail.com

Education

University of Illinois Urbana-Champaign *Aug 2024 - Present*
Ph.D. Candidate in Computer Science
Advisor: Prof. Dilek Hakkani-Tür

University of British Columbia *Sep 2019 - May 2024*
B.Sc. Honours Computer Science
Advisors: Prof. Vered Shwartz, Prof. Giuseppe Carenini

Professional Experience

Microsoft Research, Redmond, WA *Sep 2026 – Dec 2026*
Part-Time Research Scientist Intern. Mentors: Michel Galley, Philippe Laban
- Working on synthetic environments for interactive agents.

Microsoft Research, Redmond WA *May 2026 – Aug 2026*
Research Scientist Intern. Mentors: Michel Galley, Philippe Laban
- Identified user representations (natural-language descriptions used to condition simulators) as a bottleneck in user simulation, and propose hindsight distillation: an online training procedure that optimizes representation generators for downstream simulation fidelity. We train and release `UserRepGen-4B`, and submit our paper to ICLR 2027.

IBM Research, San Jose, CA *May 2025 – Aug 2025*
Research Scientist Intern, Mentors: Krishnateja Killamsetty, Marina Danilevsky
- Designed a dynamic routing system for conversational retrieval-augmented generation (RAG) pipelines that orchestrates a set of specialized LLM tools to improve response efficiency and quality.

iClinic Systems Inc., Vancouver, BC *Dec 2023 – May 2024*
Research Scientist Intern, Mentor: Vered Shwartz
- Was awarded the *Mitacs Accelerate Fellowship for Master's and PhD Students* as an undergraduate student, under the recommendation of Prof. Vered Shwartz. Worked on an information extraction pipeline tailored for clinical documents, and designed evaluations using synthetic data generation.

Amazon *Jan 2023 – Aug 2023*
Software Development Engineer Intern, Gurupa Team
- Owned the end-to-end creation of an interactive data visualization tool that streamlined the migration from Amazon's core page rendering engine. Resulted in a fully operation web application deployed to production.

Amazon *May 2022 – Aug 2022*
Software Development Engineer Intern, Dram Team
- Built a command line interface that uses cloud computing to generate and register consumer ready metadata, enabling developers to create/manage Amazon web pages.

Select Publications (Scholar Page)

[1] Learning User Representations to Improve Simulation Fidelity
Shuhaib Mehri, Philippe Laban, Michelle Galley, Dilek Hakkani-Tür, Jennifer Neville, Jianfeng Gao.
Preprint.

[2] Measuring and Mitigating the Distributional Gap Between Real and Simulated User Behaviors
Shuhaib Mehri, Philippe Laban, Sumuk Shashidhar, Marwa Abdulhai, Sergey Levine, Michel Galley, Dilek Hakkani-Tür.
NeurIPS 2026, Evaluations and Datasets Track.

[3] **PSI-Bench: Towards Clinically Grounded and Interpretable Evaluation of Depression Patient Simulators**
Nguyen Khoi Hoang, Shuhaib Mehri, Tse-An Hsu, Yi-Jyun Sun, Quynh Xuan Nguyen Truong, Khoa D Doan, Dilek Hakkani-Tür.
Social Sim Workshop at COLM 2026 (Spotlight, Best Paper Award).

[4] **MultiSessionCollab: Learning User Preferences with Memory to Improve Long-Term Collaboration**
Shuhaib Mehri, Priyanka Kargupta, Tal August, Dilek Hakkani-Tür.
EMNLP 2026 Main.

[5] **Goal Alignment in LLM-Based User Simulators for Conversational AI**
Shuhaib Mehri, Xiaocheng Yang, Takyoung Kim, Gokhan Tur, Shikib Mehri, Dilek Hakkani-Tür.
TACL 2026 (Oral at ACL 2026).

[6] **Beyond Sample-Level Feedback: Using Reference-Level Feedback to Guide Data Synthesis**
Shuhaib Mehri, Xiusi Chen, Heng Ji, Dilek Hakkani-Tür.
EACL 2026 Main (Oral).

Invited Talks

User-Centered Conversational Agents *Feb 2026*
Google Research, Host: Maximillian Chen

Learning User Preferences with Memory to Improve Long-Term Collaboration *Feb 2026*
Assistive and Inclusive Technology Workshop, Host: Mark Hasegawa-Johnson

Goal Alignment in LLM-Based User Simulators for Conversational AI *Dec 2025*
NLP Group at the University of British Columbia, Host: Vered Shwartz

Mentored Projects

[1] **When Does User Simulation Generalize? Cross-Scenario Generalization in Individual-Level User Simulation**
Shicheng Hu, Jiaxin Gao, Chi-Yu Chen, Chenhao Xu, Zhongjie Dai, Chengyu Yin, Shuhaib Mehri, Dilek Hakkani-Tür.
In Progress.

[2] **PSI-Bench: Towards Clinically Grounded and Interpretable Evaluation of Depression Patient Simulators**
Nguyen Khoi Hoang, Shuhaib Mehri, Tse-An Hsu, Yi-Jyun Sun, Quynh Xuan Nguyen Truong, Khoa D Doan, Dilek Hakkani-Tür.
Social Sim Workshop at COLM 2026 (Spotlight, Best Paper Award).

[3] **User Preference Modeling for Conversational LLM Agents: Weak Rewards from Retrieval-Augmented Interaction**
Yuren Hao, Shuhaib Mehri, ChengXiang Zhai, Dilek Hakkani-Tür.
Preprint.

Teaching

University of Illinois Urbana-Champaign *Jan 2026 – Dec 2026*

Teaching Assistant for *Advanced Topics in NLP (CS 546)*; *Algms and Data Structures for Data Science (CS 277)*

University of British Columbia *Sep 2020 – May 2024*

Teaching Assistant for *Models of Computation (CPSC 121)*; *Intro to Software Engineering (CPSC 310)*